Experiment 01:

7. Laboratory Exercise

1. pwd (Print Working Directory)

Output:

```
┌──(student㉿kali)-[~]
└─$ pwd
/home/student
```

Observation:

This command shows the folder where we are currently working. Here, it showed **/home/student**.

---

2. ls (List Directory Contents)

Output:

```
┌──(student㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  hadke  lavanya  Music  new  new1.txt  Pictures
```

```
┌──(student㉿kali)-[~/Shivam]
└─$ ls -l
total 8
-rw-rw-r-- 1 student student  0 Jul 15 12:29 new1.txt
-rw-rw-r-- 1 student student  0 Jul 17 11:41 new.txt
-rw-rw-r-- 1 student student 33 Jul 17 11:35 Radha.txt
-rw-rw-r-- 1 student student 33 Jul 17 11:34 Shivam.txt
```

```
┌──(student㉿kali)-[~/Shivam]
└─$ ls -a
.  ..  new1.txt  new.txt  Radha.txt  Shivam.txt
```

Observation:

- First command shows all the files and folders in the current folder.
- Second command shows all the files in detail. It displays information like file size, date, and permissions. Here, it showed the details of all the files in the **Shivam** folder.

- Third command shows all the files in the folder, including hidden files. Here, it displayed the hidden entries (. and ..) along with the other files in the **Shivam** folder.

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3. cd (Change Directory)

Output:

```
┌──(student㉿kali)-[~]
└─$ cd Shivam

┌──(student㉿kali)-[~/Shivam]
└─$ touch Shivam.txt Radha.txt

┌──(student㉿kali)-[~/Shivam]
└─$ ls
Radha.txt  Shivam.txt
```

Observation:

This command is used to go to another folder. Here, it changed the location to the **Shivam** folder.

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4. mkdir (Make Directory)

Output:

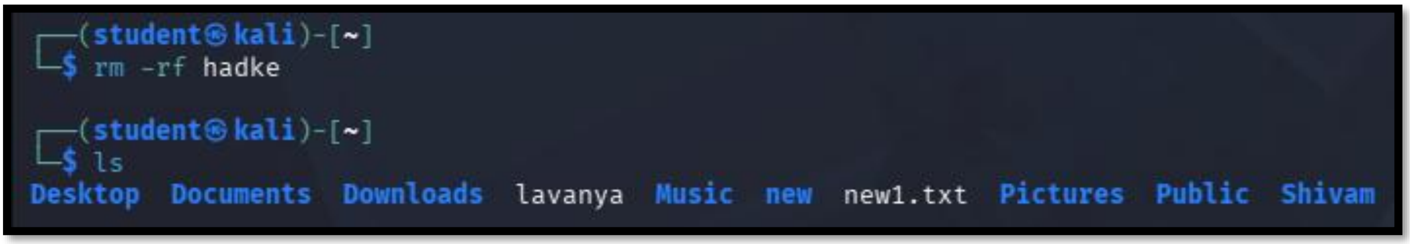
```
┌──(student㉿kali)-[~]
└─$ mkdir Shivam

┌──(student㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  hadke  lavanya  Music  new  new1.txt  Pictures  Public  Shivam
```

Observation:

This command creates a new folder. Here, a folder named **Shivam** was created.

5. rmdir / rm -rf (Remove Directory)

Output:



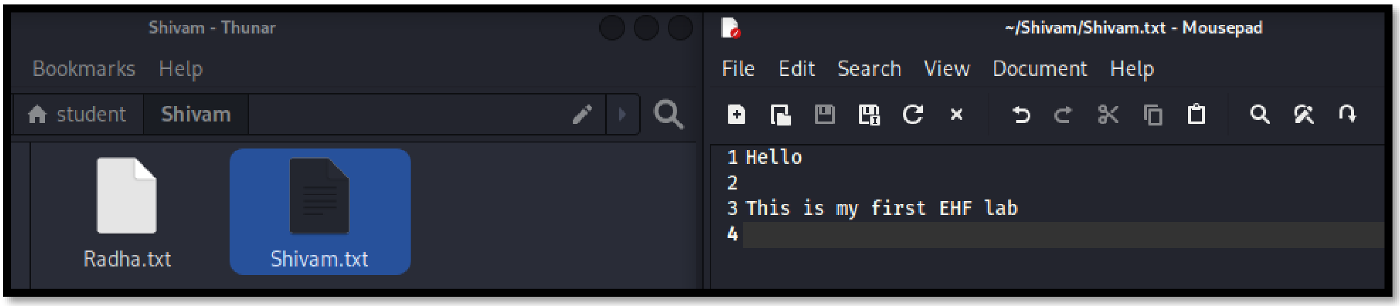
Observation:

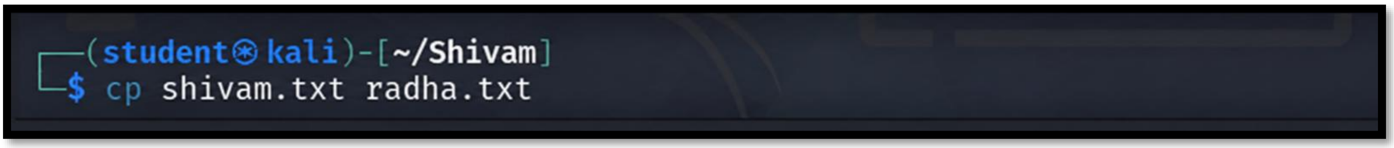
mkdir command deletes an empty folder, rm -rf deletes a non-empty folder forcefully. Here, the **hadke** folder was removed.

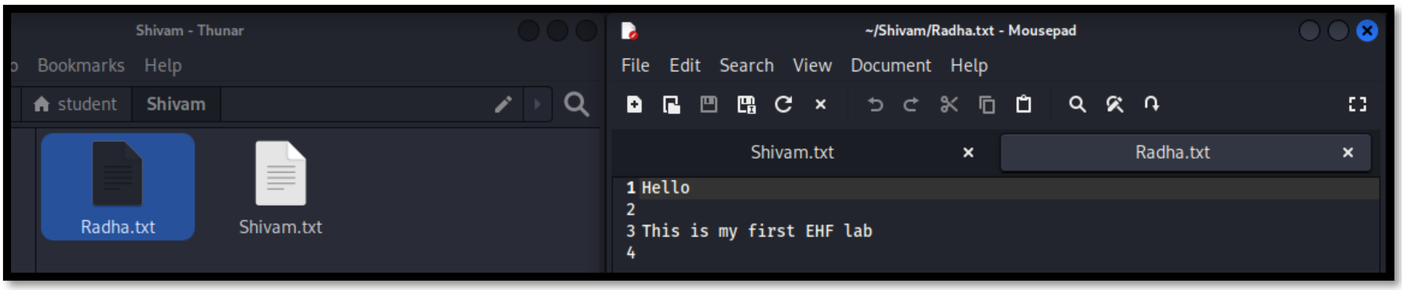
---

6. cp (Copy)

Output:





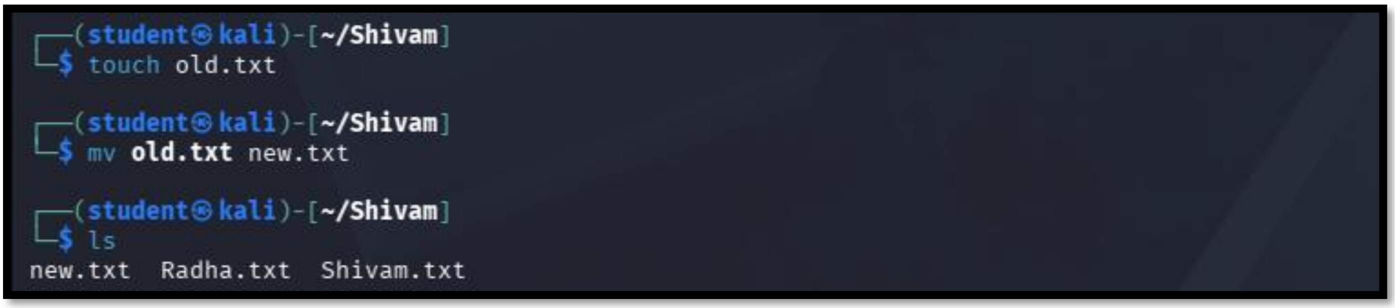


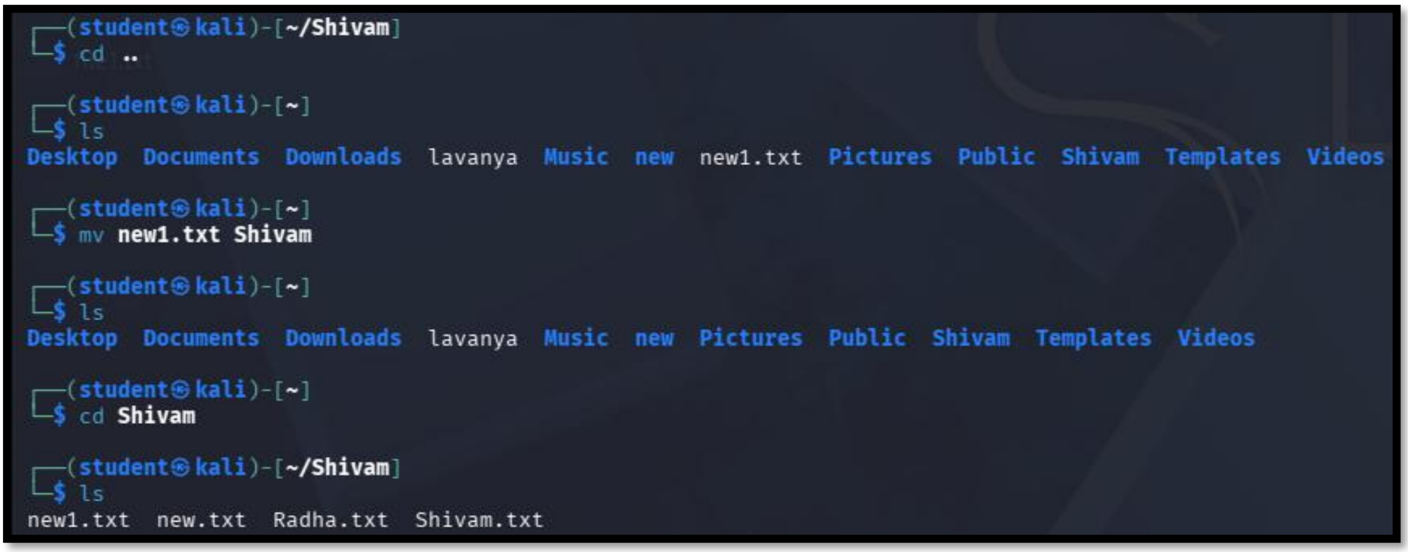
Observation:

This command copies one file to another. Here, the content of **Shivam.txt** was copied to **Radha.txt**. The original file stayed the same.

7. mv (Move / Rename)

Output:



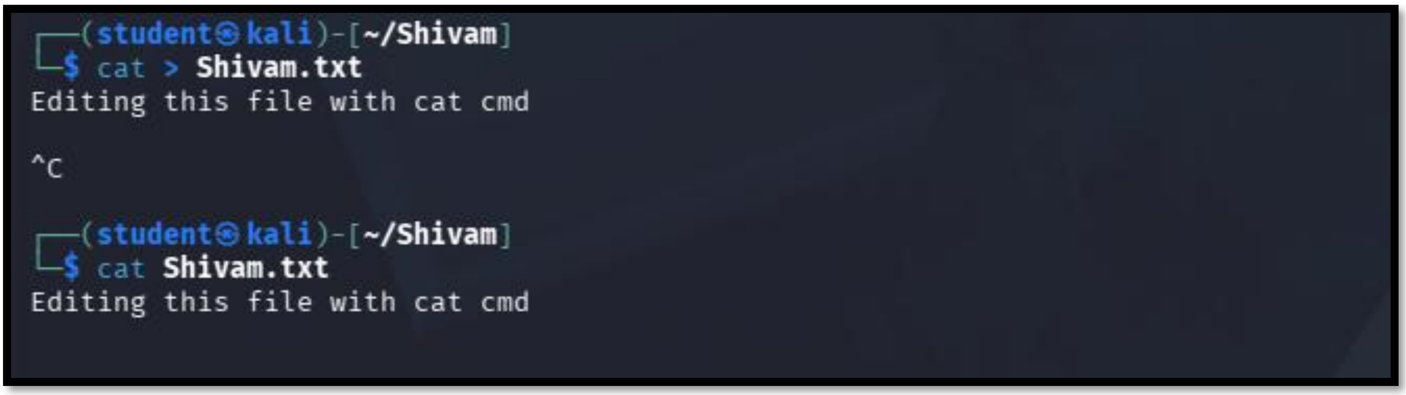


Observation:

- Command in the first screenshot was used to rename a file. Here, **old.txt** was renamed to **new.txt**. The ls command shows that **new.txt** is present and **old.txt** is no longer there.
- Command in the second screenshot moves a file from one place to another. Here, **new1.txt** was moved to the **Shivam** folder.

8. cat (Concatenate)

Output:



Observation:

- First command shows the content of a file on the terminal. Here, it displayed the text saved in **Shivam.txt**.
- This command shows the content of the Shivam.txt file on the terminal. Here, it displayed the text "**Editing this file with cat cmd**" stored in the file.

---

9. clear

Output:

Observation:

This command clears the terminal screen. It removes all the previous commands and output from the screen.

10. history

Output:

```
┌──(student㉿kali)-[~/Shivam]
└─$ history
    1  pwd
    2  ls
    3  cd Documents
    4  mkdir Ayush
```

```
   63  cd Shivam
   64  touch Shivam.txt Radha.txt
   65  ls
   66  cp Shivam.txt Radha.txt
   67  touch new.txt
   68  rm -rf new.txt
   69  touch old.txt
   70  mv old.txt new.txt
   71  ls
   72  cd ..
   73  ls
   74  mv new1.txt Shivam
   75  ls
   76  cd Shivam
   77  ls
   78  clr
   79  clear
```

Observation:

This command shows the commands that were used before.

---

11. whoami

Output:

```
┌──(student㉿kali)-[~/Shivam]
└─$ whoami
student
```

Observation:

This command shows the name of the current user. Here, the username was **student**.

12. hostname

Output:

```
┌──(student㉿kali)-[~/Shivam]
└─$ hostname
kali
```

Observation:

This command shows the name of the host. Here, the hostname was **kali**.

---

13. uname -a

Output:

```
┌──(student㉿kali)-[~/Shivam]
└─$ uname -a
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25)
```

Observation:

This command shows information about the Kali Linux system. It displayed details like the Kali Linux version and system type.

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14. date

Output:

```
┌──(student㉿kali)-[~/Shivam]
└─$ date
Friday 17 July 2026 11:51:36 AM IST
```

Observation:

This command shows the current date and time. Here, it displayed **Friday 17 July 2026, 11:51:36 AM IST**.

15. ifconfig

Output:

```
┌──(student㉿kali)-[~/Shivam]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:a00:27ff:fea4:44ea  prefixlen 64  scopeid 0x0<global>
        inet6 fd17:625c:f037:2:2353:a4cb:3809:ac3  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::a00:27ff:fea4:44ea  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:a4:44:ea  txqueuelen 1000  (Ethernet)
        RX packets 29  bytes 5789 (5.6 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 54  bytes 7206 (7.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Observation:

This command shows the network details of the system. It displayed the IP address and other network information.

---

16. ip a

Output:

```
┌──(student㉿kali)-[~/Shivam]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a4:44:ea brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 84438sec preferred_lft 84438sec
    inet6 fd17:625c:f037:2:2353:a4cb:3809:ac3/64 scope global temporary dynamic
       valid_lft 86358sec preferred_lft 14358sec
    inet6 fd17:625c:f037:2:a00:27ff:fea4:44ea/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86358sec preferred_lft 14358sec
    inet6 fe80::a00:27ff:fea4:44ea/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Observation:

This command also shows the network details. It displayed all the network connections along with their IP addresses.